MoveIt

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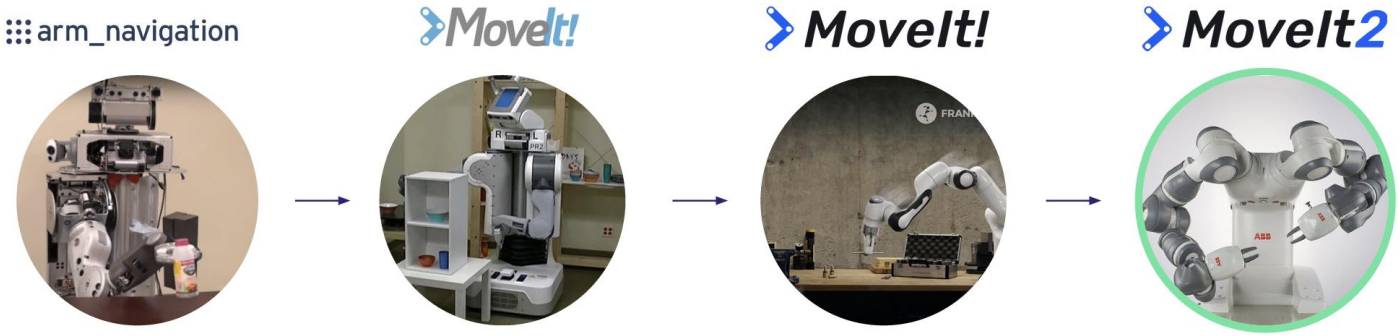
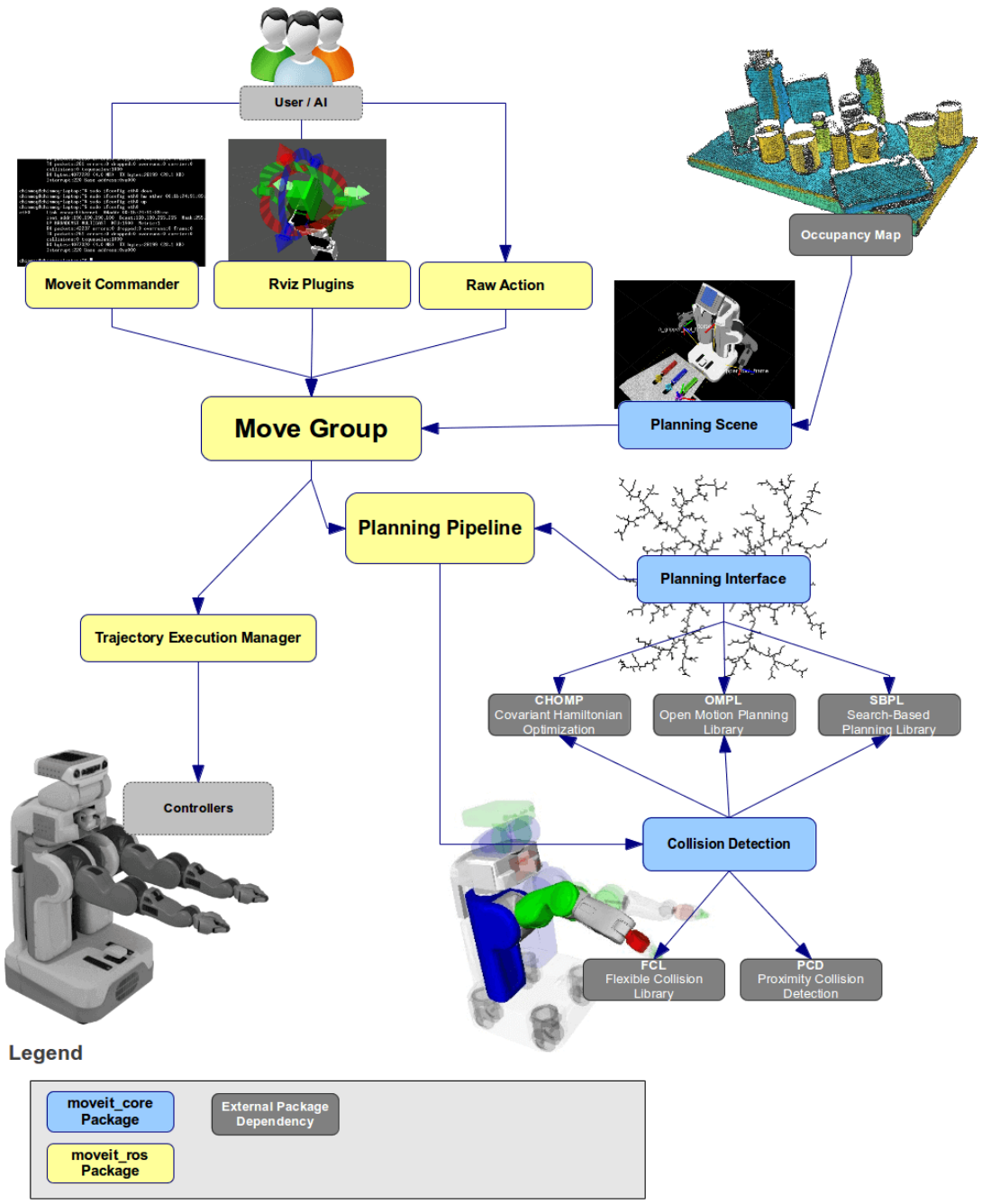
Projektiranje autonomnih sustava| Regional Center of Excellence for Robotic Technology

MoveIt

- *MoveIt2* je glavna platforma za robotsku manipulaciju u ROS 2.
- Integrira najnovija postignuća u planiranju gibanja, manipulaciji objektima, 3D percepciji, kinematici, upravljanju i navigaciji
- Prva verzija MoveIt 2 objavljena je 2019. godine
- Za starije ROS 1 sustave koristi se MoveIt 1 tutorials
- Za komercijalnu verziju dostupni su MoveIt Pro tutorials



Koncepti


User / AI
Moveit Commander
Rviz Plugins
Raw Action
Occupancy Map
Move Group
Planning Scene
Planning Pipeline
Planning Interface
Trajectory Execution Manager
CHOMP
Covariant Hamiltonian Optimization
OMPL
Open Motion Planning Library
SBPL
Search-Based Planning Library
Controllers
Collision Detection
FCL
Flexible Collision Library
PCD
Proximity Collision Detection
Legend
moveit_core Package
External Package Dependency
moveit_ros Package

Mogućnosti

Global Planners
- OMPL
- SBPL
- TrajOpt
- STOMP
- CHOMP

Cartesian Planners
- RobotState
- Descartes
- JogArm
- PilzIndustrial Motion

Inverse Kinematics
- KDL
- IKFast
- TrackIK
- LMA
- BioIK

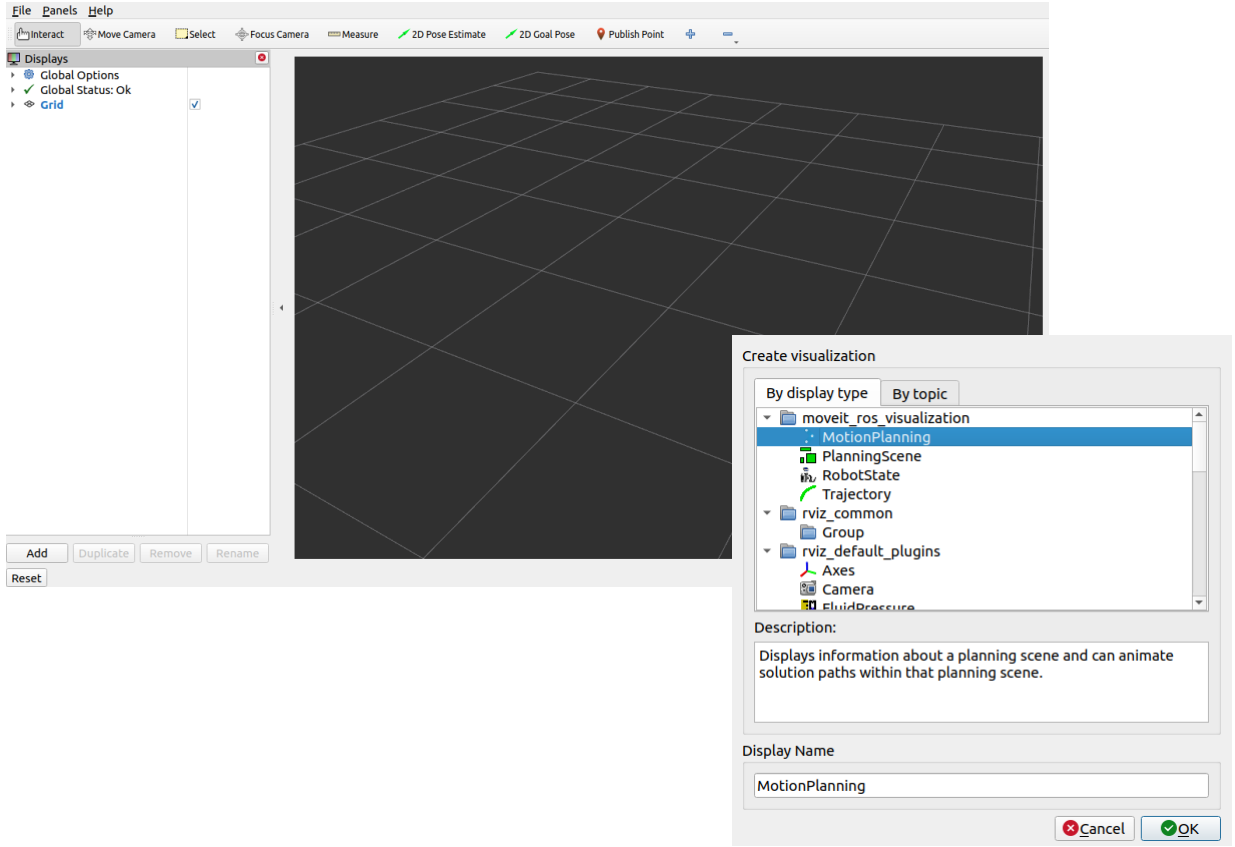
Grasping Libraries
- MoveIt Grasps
- Grasp Pose Detection (GPD)
- Intel OpenVino GPD

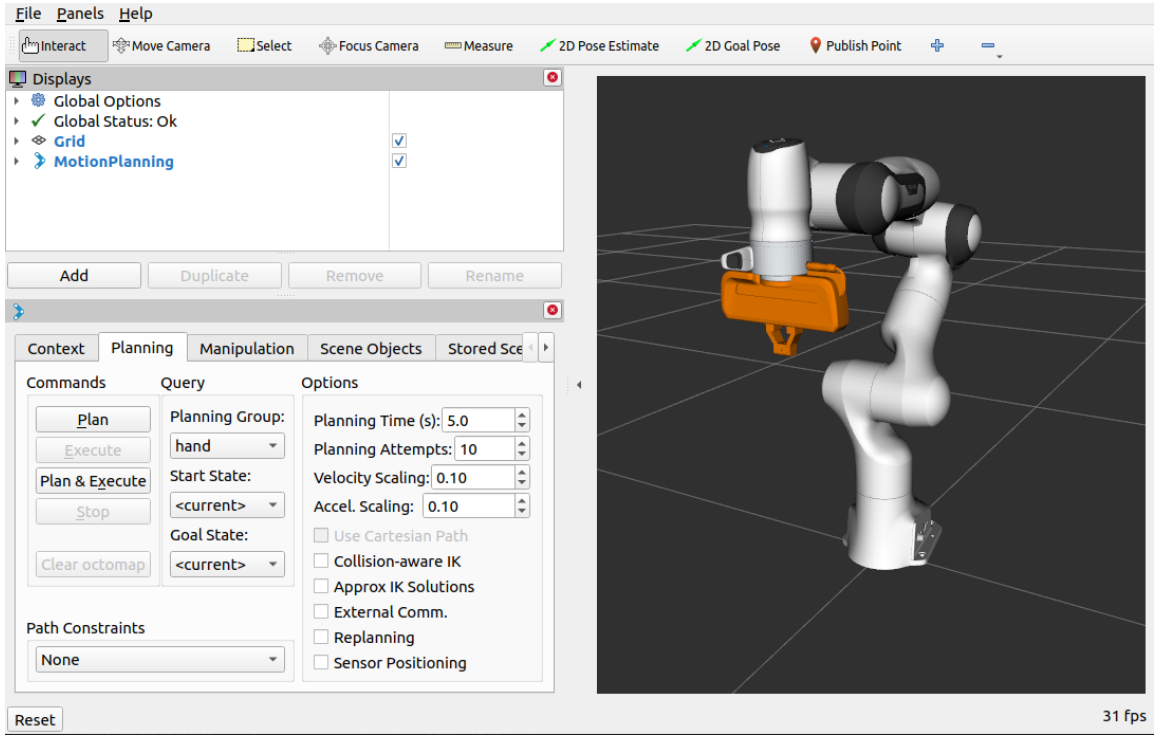
Collision Checking
- Fast Collision Library (FCL)
- Bullet

Perception / Octomap
- Depth Images
- Point Clouds

Pokretanje

```
ros2 launch moveit2_tutorials demo.launch.py rviz_config:=panda_moveit_config_demo_empty.rviz
```





Hvala!

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Konzultacije: Četvrtkom 12:00-13:00
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